

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Semester – III of B. Pharm. Examination

University Theory Examination November 2016

PH217 Pharmaceutical Chemistry-III (Organic)

Date: 18/11/2016, Friday

Time: 10:00 a.m To 01:00 p.m

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION-I

Q.1 Attempt all questions. Each question is of one marks

20

- 1 Which of the following physical properties differ for each of a pair of enantiomers?
[A] Solubility in Ethanol
[B] Direction of Rotation of Plane Polarized light
[C] Boiling Point and Melting Point
[D] Index of Refraction
- 2 Acid halides react with alcohols to give:
[A] Ester
[B] Ether
[C] Acid anhydride
[D] Amide
- 3 Which of the following is not the derivative of carboxylic acids?
[A] Ether
[B] Acid halide
[C] Ester
[D] Amide
- 4 Heat of hydrogenation and combustion of benzene is:
[A] 28.6 kcal
[B] 55.4 kcal
[C] 49.2 kcal
[D] 85.8 kcal
- 5 Chloro ethane reacts with X to form diethyl ether. What is X
[A] NaOH
[B] H₂SO₄
[C] C₂H₅ONa
[D] Na₂S₂O₃
- 6 Which of the following is not an ortho-para directing group:
[A] -NH₂
[B] -OH
[C] -OCH₃
[D] -CHO

- 7 Epoxides are compounds containing how many membered ring.
[A] 3
[B] 4
[C] 5
[D] 6
- 8 Which of the following statement regarding phenol is not correct:
[A] Phenols are stronger acid than water and alcohol
[B] Phenols are weaker acid than carboxylic acid
[C] Phenols are soluble in both NaOH and aqueous NaHCO₃
[D] Phenoxide ions are stable than the corresponding phenols
- 9 Cycloalkanes undergo which type of chemical reaction?
[A] free-radical substitution
[B] electrophilic addition
[C] free-radical addition
[D] all of above
- 10 Anchimeric Assistance word associated with the mechanism of:-
[A] SN₁
[B] SN₂
[C] Neighboring Group
[D] E₂
- 11 Carbon-carbon bond length in benzene is equal to ____ Å (approx.)
[A] 1.53
[B] 1.34
[C] 1.68
[D] 1.39
- 12 Name of the scientist who successfully gave two structures of benzene was
[A] August Kekule
[B] Dewar
[C] Huckel
[D] Both A and B
- 13 Following is the method of resolution of racemic mixture
[A] Asymmetric Transformation
[B] Chiral Synthesis
[C] Symmetric Transformation
[D] Asymmetric Transformation
- 14 Benzene undergoes which type of reaction?
[A] Electrophilic Substitution
[B] Electrophilic Addition
[C] Nucleophilic Substitution
[D] Nucleophilic Addition
- 15 Optical picture of benzene suggests following geometry
[A] Symmetrical and lying in same plane
[B] sp² Hybridized
[C] Hexagon with 120° angle
[D] All of the above
- 16 One of the following groups is activating towards benzene ring
[A] NH₂, OCH₃
[B] NO₂
[C] Cl, I
[D] None of the Above

- 17 What are Atropisomers?
 [A] Stereoisomers
 [B] Geometrical Isomers
 [C] Isomers arising out of hindered rotation about single bond
 [D] Both A and C
- 18 Which of the following is not an example of Heterogenous Catalyst?
 [A] Ni
 [B] Pt
 [C] Pd
 [D] Wilkinson's Catalyst
- 19 What is another name for Sequence Rule
 [A] CIP rule
 [B] Cahn, Ingold and Prelog sequence rule
 [C] Specification of configuration
 [D] Both A and B
- 20 Which type of isomers are of type E and Z?
 [A] Enantiomers
 [B] Geometrical Isomers
 [C] Allenes
 [D] Optical Isomers

SECTION-II

- Q.2** Attempt any **TWO** of the following
- A Give the Preparation of Carboxylic Acid. 05
- B Give any THREE reactions of Phenol along with It's Synthetic Utility. 05
- C Explain Halogenations reaction of Arenes. 05
- Q.3** Attempt any **TWO** of the following
- A Justify the Following:- 05
- 1 Carboxylic acids have higher boiling point or melting point than those of alkenes of similar size.
- 2 Aldehyde is more reactive than Ketone
- B Enumerate the any two reduction type reaction of Aldehyde and Ketone with example. 05
- C What is mean by Cycloaddition Reaction? Give 2+2 Cycloaddition reaction with it's mechanism. 05

SECTION-III

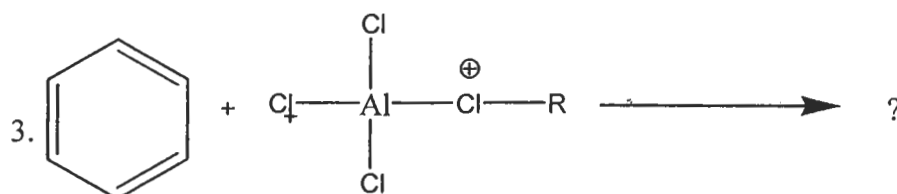
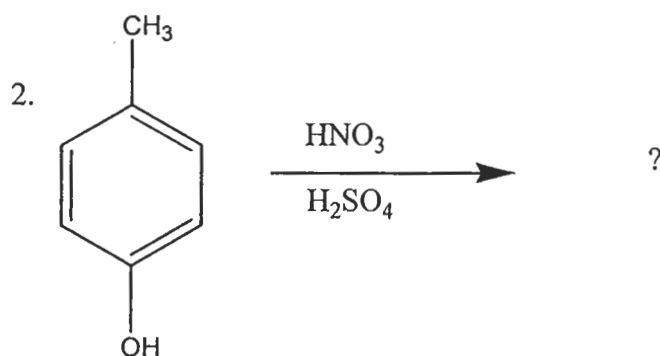
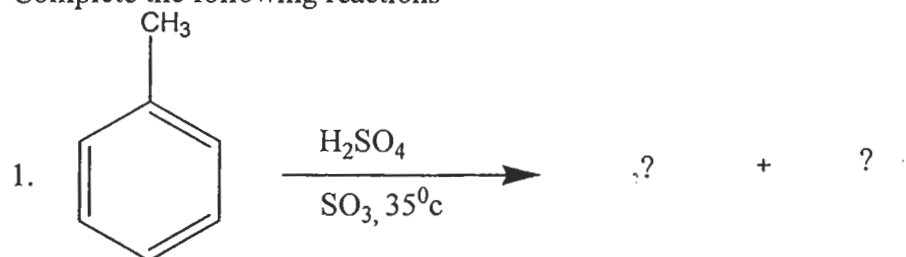
- Q.4** Attempt any **FOUR** of the following
- A Write Sequence rules for assigning Absolute Configuration with examples. 05
- B Which are the various methods for separating a racemic mixture? Explain all with examples. 05
- C Short note on Sigmatropic Reaction with an example. 05
- D Discuss Fridel-Crafts Alkylation and duality of mechanism. 05
- E Explain following terms with suitable examples; (Any Five) 05
- Relative Configuration
 - Optical Isomers
 - Chirality
 - E and Z Configuration
 - Eclipsed and Staggered
 - Erythro and Threo
- F What is Anchimeric assistance? Explain with the help of neighbouring group mechanism. 05

Q.5 Attempt any **FOUR** of the following

A Differentiate Enantiomers and Diastereoisomers with examples. **05**

B What is Bayer Stain Theory? Discuss various conformation of Cyclohexane. **05**

C Complete the following reactions **05**



4. Give structures of (i) Phenanthrene (ii) Gauche n-Butane

D What is Huckels' Rule? What properties are necessary for a compound to be aromatic? **05**

E Explain why halogens are deactivating and Ortho- Para- directors? **05**

F Discuss stereochemistry of Biphenyls. **05**